QUESTION BANK FOR IAT- ML 2026

TE AIML SEM: VI

1.200 emails that were actually spam were correctly predicted as spam. 50 emails that were actually spam were incorrectly predicted as not spam. 60 emails that were actually not spam were incorrectly predicted as spam. And 330 emails that were actually not spam were correctly predicted as not spam. For the given data build the confusion matrix and find accuracy, precision, recall and f1 score.

Ans.

	Predicted Spam	**Predicted Not Spam**
Actual Spam	TP = 200	FN = 50
Actual Not Spam	FP = 60	TN = 330

Where:

- **TP (True Positive)** = 200
- **FN (False Negative)** = 50
- **FP (False Positive)** = 60
- **TN (True Negative)** = 330

Calculations

1. Accuracy = (TP + TN) / (TP + FP + FN + TN)
= (200 + 330) / (200 + 60 + 50 + 330)
= 530 / 640
≈ **0.8281** or **82.81%**

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2. Precision = TP / (TP + FP)
= 200 / (200 + 60)
= 200 / 260
≈ **0.7692** or **76.92%**

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3. Recall = TP / (TP + FN)
= 200 / (200 + 50)
= 200 / 250
= **0.8** or **80%**

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4. F1 Score = 2 * (Precision * Recall) / (Precision + Recall)
= 2 * (0.7692 * 0.8) / (0.7692 + 0.8)
≈ 2 * 0.6154 / 1.5692
≈ **0.7832** or **78.32%**

Final Results

- **Accuracy**: 82.81%
- **Precision**: 76.92%
- **Recall**: 80%
- **F1 Score**: 78.32%

2.In a SVM explain the kernel trick.

Ans. The kernel trick is a method used in SVMs to enable them to classify non-linear data using a linear classifier. By applying a kernel function, SVMs can implicitly map input data into a higher-dimensional space where a linear separator (hyperplane) can be used to divide the classes. This mapping is computationally efficient because it avoids the direct calculation of the coordinates in this higher space.

Types of Kernel Functions

Several kernel functions can be used, each suited to different types of data distributions:

- **Linear Kernel**: No mapping is needed as the data is already assumed to be linearly separable.
- **Polynomial Kernel**: Maps inputs into a polynomial feature space, enhancing the classifier's ability to capture interactions between features.
- **Radial Basis Function (RBF) Kernel**: Also known as the Gaussian kernel, it is useful for capturing complex regions by considering the distance between points in the input space.
- **Sigmoid Kernel**: Mimics the behavior of neural networks by using a sigmoid function as the kernel.

How Does the Kernel Trick Work?

The kernel trick relies on the inner products of vectors. For SVMs, the decision function is based on the dot products of vectors within the input space. Kernel functions replace these dot products with a non-linear function that computes a dot product in a higher-dimensional space. Importantly, the computation of this dot product via the kernel function does not require explicit knowledge of the coordinates in the higher space, thus saving computational resources and time.

The kernel trick is typically expressed as:

$K(x,y)=\phi(x)\cdot\phi(y)$ $K(x,y)=\phi(x)\cdot\phi(y)$

where,

- x and y are two vectors in the original input space
- ϕ ϕ is the mapping function to the higher-dimensional space.

3. What is Machine Learning? What are the steps in developing a machine learning application?

Machine Learning (ML)

Machine Learning is a subfield of artificial intelligence where **systems learn patterns from data** and **make decisions or predictions** without being explicitly programmed.

Definition

Machine Learning is the process by which computers learn from historical data to **predict outcomes**, **classify information**, or **make decisions** automatically.

Examples:

- Spam email detection
- Loan approval prediction
- Image recognition

Steps of developing a Machine Learning Application:

1.Problem Definition

- Clearly define the business or real-world problem.
- Decide whether ML is appropriate and what type (classification,
- regression, clustering, etc.).

2.Data Collection

- Gather relevant data from databases, APIs, sensors, logs, or web scraping.

- Ensure data quantity and quality are sufficient.

3.Data Exploration & Understanding

- Analyze data structure, distributions, correlations, and patterns.
- Identify missing values, outliers, and biases.
- **4.Data Preprocessing**
- Clean data (handle missing values, duplicates, noise).
- Normalize/scale features.
- Encode categorical variables.
- Split data into training, validation, and test sets.

5.Feature Engineering

- Select relevant features.
- Create new features from existing data.
- Reduce dimensionality if needed.

6.Model Selection

- Choose suitable algorithms (e.g., Linear Regression, Decision Trees, Neural Networks).
- Consider performance, interpretability, and computational cost.

7.Model Training

- Train the model using the training dataset.
- Tune hyperparameters.

8.Model Evaluation

- Evaluate using appropriate metrics (accuracy, precision, recall, F1-score, RMSE, etc.).
- Validate performance on unseen data.
- 9.**Model Optimization**
- Improve performance through feature refinement, algorithm changes, or hyperparameter tuning.
- Address overfitting or underfitting.
- **10.Deployment**

- Integrate the model into an application (web app, mobile app, API).
- Use tools like Flask, FastAPI, Docker, or cloud platforms.
- 11.**Monitoring & Maintenance**
- Monitor model performance in real-world use.
- Detect data drift or performance degradation.
- Retrain or update the model as needed.
- 12.**Documentation & Reporting**
- Document assumptions, data sources, model behavior, and limitations.
- Communicate results to stakeholders.

4. What is supervised and unsupervised learning.

Ans. **1.Supervised Learning**

Definition:

- Supervised learning is defined as when a model gets trained on a "Labelled Dataset". Labelled datasets have both input and output parameters.
- In Supervised Learning algorithms learn to map points between inputs and correct outputs. It has both training and validation datasets labelled.

Learning with **labeled input data** (each input has a known output).

Goal:

Predict outcomes for new/unseen data.

Examples:

- Classification (Spam/Not Spam)
- Regression (House Price Prediction)

2.Unsupervised Learning

Definition:

- Unsupervised learning is a type of machine learning technique in which an **algorithm discovers patterns and relationships** using **unlabeled data**.
- Unlike supervised learning, unsupervised learning doesn't involve providing the algorithm with labeled target outputs.

- The primary goal of Unsupervised learning is often to discover hidden patterns, similarities, or clusters within the data, which can then be used for various purposes, such as data exploration, visualization, dimensionality reduction, and more.

Learning from **unlabeled input data** (no predefined output).

Goal:

Find hidden patterns or structure in data.

Examples:

- Clustering (K-Means)
- Association Rules (Apriori/FP-Growth)

5. Explain Overfitting and Underfitting with example.

Ans. Overfitting:

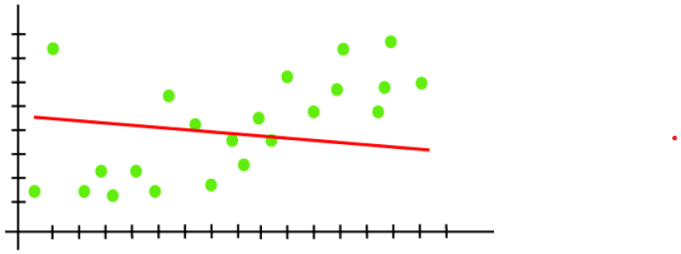
- Overfitting occurs when our machine learning model tries to cover all the data points or more than the required data points present in the given dataset.
- Because of this, the model starts catching noise and inaccurate values present in the dataset, and all these factors reduce the efficiency and accuracy of the model. The overfitted model has **low bias** and **high variance.**
- Let's understand with a simple example where we have a few training data sets such as 1000 mangoes, 1000 apples, 1000 bananas, and 5000 papayas.
- Then there is a considerable probability of identification of an apple as papaya because we have a massive amount of biased data in the training data set; hence prediction got negatively affected.

Underfitting:

- Underfitting occurs when our machine learning model is not able to capture the underlying trend of the data.
- To avoid the overfitting in the model, the fed of training data can be stopped at an early stage, due to which the model may not learn enough from the training data.
- As a result, it may fail to find the best fit of the dominant trend in the data.

- In the case of underfitting, the model is not able to learn enough from the training data, and hence it reduces the accuracy and produces unreliable predictions.
- An underfitted model has high bias and low variance.

Example: We can understand the underfitting using below output of the linear regression model:



As we can see from the above diagram, the model is unable to capture the data points present in the plot.
How to avoid underfitting:

- By increasing the training time of the model.
- By increasing the number of features.

6. Explain Eigen Values and Vectors.

- Ans. Eigenvalues and Eigenvectors are the scalar and vector quantities associated with Matrix used for linear transformation.
- The word 'Eigen' is of German Origin which means 'characteristic'. Hence, these are the characteristic value that indicates the factor by which eigenvectors are stretched in their direction.
- It doesn't involve the change in the direction of the vector except when the eigenvalue is negative.
- When the eigenvalue is negative the direction is just reversed.
- The equation for eigenvalue is given by

$$Av = \lambda v$$

where A is the matrix

v is associated eigenvector

λ is scalar eigenvalue

- Eigenvectors for square matrices are defined as non-zero vector values which when multiplied by the square matrices give the scaler multiple of the vector, i.e. we define an eigenvector for matrix A to be "v" if it specifies the condition, Av = λv

- The scaler multiple λ in the above case is called the eigenvalue of the square matrix.
- We always have to find the eigenvalues of the square matrix first before finding the eigenvectors of the matrix.
- The eigenvector equation is,

$$Av = \lambda v$$

where,

A is the given square matrix

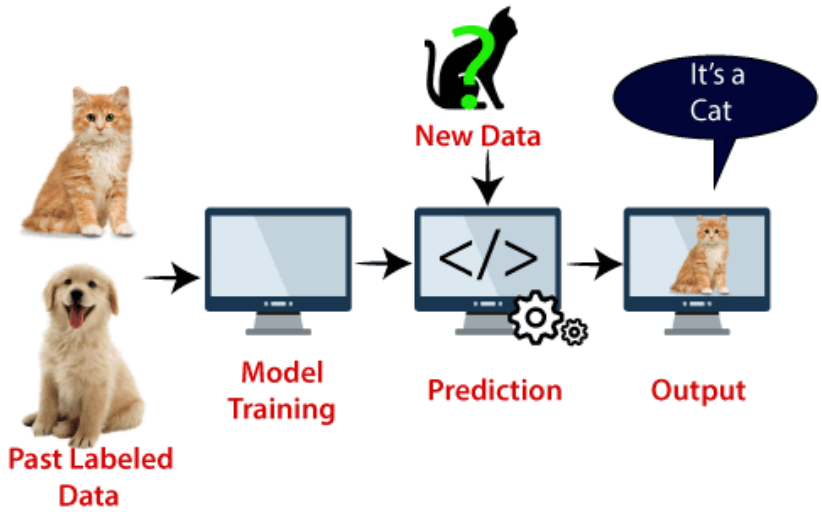
v is the eigenvector of matrix A

λ is any scaler multiple

7.Discuss Support Vector Machine.

- Ans. Support Vector Machine (SVM) is a powerful machine learning algorithm used for linear or nonlinear classification, regression, and even outlier detection tasks.
- SVMs can be used for a variety of tasks, such as text classification, image classification, email spam detection, handwriting identification, face detection, and anomaly detection.
- SVMs are adaptable and efficient in a variety of applications because they can manage high-dimensional data and nonlinear relationships.
- SVM algorithms are very effective as we try to find the maximum separating hyperplane between the different classes available in the target feature.
- SVM used for both classification and regression. Though we say regression problems as well it's best suited for classification.
- Suppose we see a strange cat that also has some features of dogs, so if we want a model that can accurately identify whether it is a cat or dog, so such a model can be created by using the SVM algorithm.
- We will first train our model with lots of images of cats and dogs so that it can learn about different features of cats and dogs, and then we test it with this strange creature.
- So as **support vector** creates a decision boundary between these two data (cat and dog) and choose extreme cases (support vectors), it will see the extreme case of cat and dog.

- On the basis of the support vectors, it will classify it as a cat. Consider the below diagram:



8. Explain SVD and its application.

Ans. Singular Value Decomposition (SVD) is a factorization method in linear algebra that decomposes a matrix into three other matrices, providing a way to represent data in terms of its singular values.

SVD helps you split that table into three parts:

- **U**: This part tells you about the people (like their general preferences).
- **Σ**: This part shows how important each factor is (how much each rating matters).
- **V^T**: This part tells you about the products (how similar they are to each other)

Name	Movie 1 Rating	Movie 2 Rating
Amit	5	3
Sanket	4	2
Harsh	2	5

- SVD breaks this table into three smaller parts: one that shows people's preferences, one that shows the importance of each movie, and one that shows how similar the movies are to each other
- Mathematically, the SVD of a matrix A (of size $m \times n$) is represented as: $A = U\Sigma V^T$

Here:

- U: An $m \times m$ orthogonal matrix whose columns are the left singular vectors of A.
- Σ: A diagonal $m \times n$ matrix containing the singular values of A in descending order.
- V^T: The transpose of an $n \times n$ orthogonal matrix, where the columns are the right singular vectors of A.

Applications in Digital Signal Processing (DSP) and Image Processing

- **Digital Signal Processing**: SVD can be used to analyze signals and filter noise.
- **Image Processing**: SVD is used for image compression and denoising. It helps in reducing the dimensionality of image data by preserving the most significant singular values and discarding the rest

9.What is trace of matrix. What are its properties?

Ans.

The trace of a square matrix is **the sum of its main diagonal elements** $a_{11} + a_{22} + \cdots + a_{nn}$, denoted as $\mathrm{tr}(A)$.
It represents a linear mapping that is invariant under cyclic permutations and transposition, serving as a key scalar invariant in linear algebra equal to the sum of the matrix's eigenvalues.

Key Properties of the Trace of a Matrix

- **Linearity:** The trace of a sum is the sum of traces, and the trace of a scalar multiple is the scalar multiple of the trace:
  $\mathrm{tr}(A + B) = \mathrm{tr}(A) + \mathrm{tr}(B)$ and
  $\mathrm{tr}(cA) = c\,\mathrm{tr}(A)$.
- **Transpose Invariance:** The trace of a matrix is equal to the trace of its transpose:
  $\mathrm{tr}(A) = \mathrm{tr}(A^T)$.
- **Cyclic Property:** For matrices A and B, the trace of their product is commutative, meaning
  $\mathrm{tr}(AB) = \mathrm{tr}(BA)$.
  Consequently,
  $\mathrm{tr}(ABC) = \mathrm{tr}(BCA) = \mathrm{tr}(CAB)$.
- **Eigenvalue Sum:** The trace is equal to the sum of the eigenvalues of a matrix, counted with their multiplicities.
- **Similar Matrices:** Similar matrices have the same trace:
  $\mathrm{tr}(P^{-1}AP) = \mathrm{tr}(A)$.
- **Identity Matrix:** The trace of an $n \times n$ identity matrix I is n:
  $\mathrm{tr}(I) = n$.

10.Find Singular Value Decomposition of given Matrix and indicate insights of about linear transformation conveyed by this method.

$$A = \begin{bmatrix} 1 & 1 \\ 7 & 7 \end{bmatrix}$$

Soln. :

$$\text{Let } A = \begin{bmatrix} 1 & 1 \\ 7 & 7 \end{bmatrix}$$

$$\text{Transpose A to get } A^T = \begin{bmatrix} 1 & 7 \\ 1 & 7 \end{bmatrix}$$

Calculate $A \times A^T$

$$A \times A^T = \begin{bmatrix} 1 & 1 \\ 7 & 7 \end{bmatrix} \times \begin{bmatrix} 1 & 7 \\ 1 & 7 \end{bmatrix} = \begin{bmatrix} 2 & 14 \\ 14 & 98 \end{bmatrix}$$

Calculate $A^T \times A$

$$A^T \times A = \begin{bmatrix} 1 & 7 \\ 1 & 7 \end{bmatrix} \times \begin{bmatrix} 1 & 1 \\ 7 & 7 \end{bmatrix} = \begin{bmatrix} 50 & 50 \\ 50 & 50 \end{bmatrix}$$

To find the value of left singular vector U, let's find the eigen values and eigenvectors of $A \times A^T = \begin{bmatrix} 2 & 14 \\ 14 & 98 \end{bmatrix}$

$$\begin{bmatrix} 2-\lambda & 14 \\ 14 & 98-\lambda \end{bmatrix} = 0$$

$$(2-\lambda)(98-\lambda) - 14 \times 14 = 0$$

$$\lambda^2 - 100\lambda + 196 - 196 = 0$$

$$\lambda^2 - 100\lambda = 0$$

Hence $\lambda = 0$ or $\lambda = 100$

Let's find the 1st eigenvector corresponding to $\lambda = 0$. Let's call it $v = \begin{bmatrix} x \\ y \end{bmatrix}$.

$$\begin{bmatrix} 2 & 14 \\ 14 & 98 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = 0 \times \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{bmatrix} 2x + 14y \\ 14x + 98y \end{bmatrix} = 0$$

$$\text{Hence, } x = -7y$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -7 \\ 1 \end{bmatrix}$$

To make the length of this eigenvector as 1, divide the eigenvector with its current length

$$\sqrt{(-7)^2 + (1)^2} = \sqrt{50} = 7.07$$

Hence, 1st eigenvector of unit length is

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -7/7.07 \\ 1/7.07 \end{bmatrix} = \begin{bmatrix} -0.99 \\ 0.14 \end{bmatrix}$$

Let's find the 2nd eigenvector corresponding to $\lambda = 100$. Let's call it $v = \begin{bmatrix} x \\ y \end{bmatrix}$.

$$\begin{bmatrix} 2 & 14 \\ 14 & 98 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = 100 \times \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{bmatrix} 2x + 14y \\ 14x + 98y \end{bmatrix} - \begin{bmatrix} 100x \\ 100y \end{bmatrix} = 0$$

$$\begin{bmatrix} -98x + 14y \\ 14x - 2y \end{bmatrix} = 0$$

Ans

Hence, $x = \frac{y}{7}$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 7 \end{bmatrix}$$

To make the length of this eigenvector as 1, divide the eigenvector with its current length

$$\sqrt{(1)^2 + (7)^2} = \sqrt{50} = 7.07.$$

Hence, 2nd eigenvector of unit length is

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1/7.07 \\ 7/7.07 \end{bmatrix} = \begin{bmatrix} 0.14 \\ 0.99 \end{bmatrix}$$

You sort the eigenvectors according to eigen values.

Hence,

$$U = \begin{bmatrix} 0.14 & -0.99 \\ 0.99 & 0.14 \end{bmatrix}$$

To find the value of right singular vector V^T, let's find the eigenvalues and eigenvectors of $A^T \times A = \begin{bmatrix} 50 & 50 \\ 50 & 50 \end{bmatrix}$

$$\begin{bmatrix} 50-\lambda & 50 \\ 50 & 50-\lambda \end{bmatrix} = 0$$

$$(50-\lambda)(50-\lambda) - 50 \times 50 = 0$$

$$\lambda^2 - 100\lambda + 2500 - 2500 = 0$$

$$\lambda^2 - 100\lambda = 0$$

Hence $\lambda = 0$ or $\lambda = 100$

Let's find the 1st eigenvector corresponding to $\lambda = 0$. Let's call it $v = \begin{bmatrix} x \\ y \end{bmatrix}$.

$$\begin{bmatrix} 50 & 50 \\ 50 & 50 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = 0 \times \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{bmatrix} 50x + 50y \\ 50x + 50y \end{bmatrix} = 0$$

Hence, $x = -y$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

To make the length of this eigenvector as 1, divide the eigenvector with its current length

$$\sqrt{(-1)^2 + (1)^2} = \sqrt{2} = 1.41.$$

Hence, 1st eigenvector of unit length is

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -1/1.41 \\ 1/1.41 \end{bmatrix} = \begin{bmatrix} -0.70 \\ 0.70 \end{bmatrix}$$

Let's find the 2nd eigenvector corresponding to $\lambda = 100$. Let's call it $v = \begin{bmatrix} x \\ y \end{bmatrix}$.

$$\begin{bmatrix} 50 & 50 \\ 50 & 50 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = 100 \times \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{bmatrix} 50x + 50y \\ 50x + 50y \end{bmatrix} - \begin{bmatrix} 100x \\ 100y \end{bmatrix} = 0$$

$$\begin{bmatrix} -50x + 50y \\ 50x - 50y \end{bmatrix} = 0$$

Hence, $x = y$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

To make the length of this eigenvector as 1, divide the eigenvector with its current length

$$\sqrt{(1)^2 + (1)^2} = \sqrt{2} = 1.41.$$

Hence, 2nd eigenvector of unit length is

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1/1.41 \\ 1/1.41 \end{bmatrix} = \begin{bmatrix} 0.70 \\ 0.70 \end{bmatrix}$$

You sort the eigenvectors according to eigenvalues.

Hence,

$$V^T = \begin{bmatrix} 0.70 & -0.70 \\ 0.70 & 0.70 \end{bmatrix}$$

Singular values are square root of eigenvalues and are sorted in descending order. It is a matrix of m × n where all elements are 0 except the diagonal elements.

$$\varepsilon = \begin{bmatrix} \sqrt{100} & 0 \\ 0 & \sqrt{0} \end{bmatrix} = \begin{bmatrix} 10 & 0 \\ 0 & 0 \end{bmatrix}$$

Hence, SVD of $\begin{bmatrix} 1 & 1 \\ 7 & 7 \end{bmatrix}$ can be written as following.

$$\begin{bmatrix} 1 & 1 \\ 7 & 7 \end{bmatrix} = U.\varepsilon V^T = \begin{bmatrix} 0.14 & -0.99 \\ 0.99 & 0.14 \end{bmatrix} \times \begin{bmatrix} 10 & 0 \\ 0 & 0 \end{bmatrix} \times \begin{bmatrix} 0.70 & -0.70 \\ 0.70 & 0.70 \end{bmatrix}$$

Just for fun and confirmation, if you go ahead and multiply the matrices you got after decomposition, you will get the original matrix (very close values).

So, $\begin{bmatrix} 0.14 & -0.99 \\ 0.99 & 0.14 \end{bmatrix} \times \begin{bmatrix} 10 & 0 \\ 0 & 0 \end{bmatrix} \times \begin{bmatrix} 0.70 & -0.70 \\ 0.70 & 0.70 \end{bmatrix}$ will give you $\begin{bmatrix} 0.98 & -0.98 \\ 6.93 & -6.93 \end{bmatrix}$ which is very close to the matrix that you started with $\begin{bmatrix} 1 & 1 \\ 7 & 7 \end{bmatrix}$

Note here that signs of matrices U and V do not matter. You may get different signs. As long as the absolute values are correct, it is a correct SVD.

11. Diagonalize the matrix A

$$A = \begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix}$$

Ans. **Step 1: Find the eigenvalues**

Compute the characteristic polynomial:

$$\begin{aligned} \det(A - \lambda I) &= \begin{vmatrix} 2-\lambda & 2 \\ 1 & 3-\lambda \end{vmatrix} \\ &= (2-\lambda)(3-\lambda) - 2 \\ &= 6 - 5\lambda + \lambda^2 - 2 \\ &= \lambda^2 - 5\lambda + 4 \end{aligned}$$

Solve:

$$\begin{aligned} &\lambda^2 - 5\lambda + 4 = 0 \\ &(\lambda - 4)(\lambda - 1) = 0 \\ &\lambda_1 = 4, \lambda_2 = 1 \end{aligned}$$

Step 2: Find eigenvectors

For $\lambda = 4$:

$$A - 4I = \begin{pmatrix} -2 & 2 \\ 1 & -1 \end{pmatrix}$$

Solve:

$$-2x + 2y = 0$$
$$x = y$$

Eigenvector:

$$v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

For $\lambda = 1$:

$$A - I = \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}$$

Solve:

$$x + 2y = 0$$
$$x = -2y$$

Eigenvector:

$$v_2 = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

Step 3: Form $P$and D

Let

$$P = \begin{pmatrix} 1 & -2 \\ 1 & 1 \end{pmatrix}$$
$$D = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}$$

Final Answer

$$A = PDP^{-1}$$

where

$$P = \begin{pmatrix} 1 & -2 \\ 1 & 1 \end{pmatrix}, D = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}$$

tep 1: Compute the determinant of P

$$\det(P) = \begin{vmatrix} 1 & -2 \\ 1 & 1 \end{vmatrix} = (1)(1) - (-2)(1)$$
$$= 1 + 2 = 3$$

Step 2: Use the 2×2 inverse formula

For a 2×2matrix

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Apply it:

$$P^{-1} = \frac{1}{3}\begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix}$$

Final Answer

$$\boxed{P^{-1} = \frac{1}{3}\begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix}}$$

Since there are two distinct eigenvalues, **A is diagonalizable**.

12.Find the values of a and b to estimate the linear regression equation (y= a+bx) for the following two sets of data:

X	2	4	6	8
Y	3	7	5	10

To estimate the linear regression equation:

y= ax + b

Ans.

Step 1: Compute Sums

Let's calculate:

- $n = 4$
- $\sum x$
- $\sum y$
- $\sum xy$
- $\sum x^2$

X	Y	XY	X^2
2	3	6	4
4	7	28	16
6	5	30	36
8	10	80	64
∑		144	120
	25		

∑x = 20

$$b = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$$

$$a = \frac{\sum y - b\sum x}{n}$$

Slope b:

$$b = \frac{4(144) - (20)(25)}{4(120) - (20)^2} = \frac{576 - 500}{480 - 400} = \frac{76}{80} = 0.95$$

Intercept a:

$$a = \frac{25 - 0.95(20)}{4} = \frac{25 - 19}{4} = \frac{6}{4} = 1.5$$

- $a = 1.5$
- $b = 0.95$

$$\boxed{y = 1.5 + 0.95x}$$

13.We need to compute the slope b and intercept a using the least squares method.Find a linear regression equation for the following two sets of data:

Time X (second)	Mass(y)
5	40
7	120
12	180
16	210
20	240

Ans. We'll find the **least squares regression line**

$$y = a + bx$$

where

- b= slope
- a= intercept

Step 1: Organize the Data

x	y	x^2	xy
5	40	25	200
7	120	49	840
12	180	144	2160
16	210	256	3360
20	240	400	4800

Now compute sums:

$$\sum x = 5 + 7 + 12 + 16 + 20 = 60$$
$$\sum y = 40 + 120 + 180 + 210 + 240 = 790$$
$$\sum x^2 = 25 + 49 + 144 + 256 + 400 = 874$$
$$\sum xy = 200 + 840 + 2160 + 3360 + 4800 = 11360$$

Number of data points:

$$n = 5$$

Step 2: Compute the Slope b

$$b = \frac{n\sum xy - (\sum x)(\sum y)}{n\sum x^2 - (\sum x)^2}$$

Substitute values:

$$b = \frac{5(11360) - (60)(790)}{5(874) - 60^2}$$
$$b = \frac{56800 - 47400}{4370 - 3600}$$
$$b = \frac{9400}{770}$$
$$b \approx 12.21$$

Step 3: Compute the Intercept a

$$a = \frac{\sum y - b\sum x}{n}$$
$$a = \frac{790 - (12.21)(60)}{5}$$
$$a = \frac{790 - 732.6}{5}$$
$$a = \frac{57.4}{5}$$
$$a \approx 11.48$$

Final Linear Regression Equation

$$\boxed{y = 11.48 + 12.21x}$$

14.Differentiate between Lasso and Ridge Regression.

Ans.

Differences between Ridge and Lasso Regression

Ridge regression	Lasso regression
Ridge regression is mostly used to reduce the overfitting in the model, and it includes all the features present in the model. It reduces the complexity of the model by shrinking the coefficients.	Lasso regression helps to reduce the overfitting in the model as well as feature selection.
Ridge regression never sets the value of coefficient to absolute zero.	Lasso Regression tends to make coefficients to absolute zero.
For ridge regression, the total sum of squares of coefficients is less than or equal to s. Here, s is a constant which exists for each value of the shrinkage factor λ.	for Lasso regression, the total sum of modulus of coefficients is less than or equal to s. Here, s is a constant which exists for each value of the shrinkage factor λ.
the ridge regression is described by $\beta1^2 + \beta2^2 \leq s$.	the lasso regression equation becomes, $\lvert\beta1\rvert + \lvert\beta2\rvert \leq s$.